Liebherr Group – Speaker notes

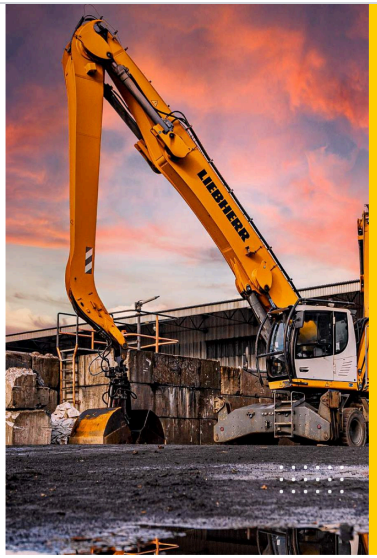
HAT1 Financial Reporting · 12 talk slides + 17 backup slides (13–29) · planned speaking time 9:40 min · 1245 words of notes (≈ 125 words per minute, calm B2 pace)

Red line: **Should a long-term investor take a minority stake in Liebherr – and if not, why not yet?**
Answer first (slide 2) → why: company and model (3) → performance (4) → can we trust the figures? (5–6) → position, cash and working capital (7–8) → how the workbook supports it (9) → conditions for reassessment (10) → conclusion (11).

All figures are taken from the report and the Excel workbook. Actual adjustments (Caterpillar LIFO → FIFO, 2022 goodwill reclassification) are kept separate from illustrative sensitivities (provision charge, depreciation rate).

01 Title

0:25



HAT1 FINANCIAL REPORTING · 2026

Liebherr Group

Financial Reporting and Investment Analysis 2022–2025

CENTRAL QUESTION

How financially strong is Liebherr, and which factors will drive its future performance?

For a long-term investor: take a minority stake now – or not yet?

Benchmark: Caterpillar Inc. and the average of 105 listed US machinery companies

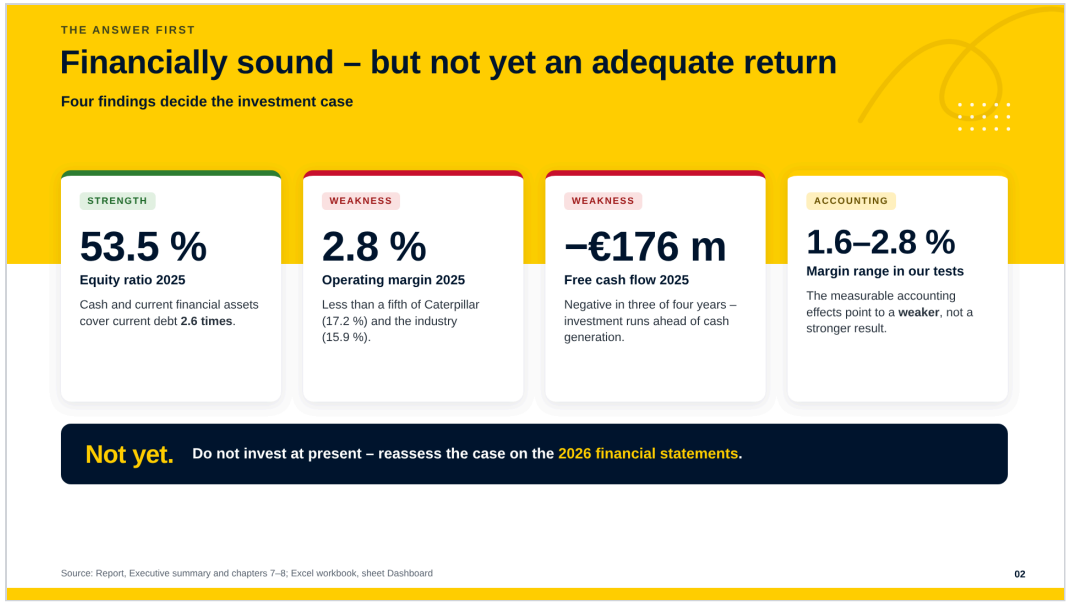
PRESENTED BY

Simon Beermann · Adam Fliegel · Zofia Narkiewicz · Philipp Hoffmann · Zarah Gilmore Islam

Fontys University of Applied Sciences, Venlo · International Finance and Control · Lecturer: Kim Verspeek

Good morning everyone. We are Simon, Adam, Zofia, Philipp and Zarah, and we present our analysis of the Liebherr Group for 2022 to 2025.

Our report is written for a long-term investor who thinks about buying a minority stake. Our central question is: how financially strong is Liebherr, and which factors will drive its future performance? For the investor, this leads to one decision: take a minority stake now – or not yet?



We give you the answer first: not yet.

Liebherr is financially sound. Equity finances 53.5 per cent of total assets, and cash covers current debt 2.6 times, so the risk of financial distress is low.

But a strong balance sheet is not a good investment by itself. The operating margin was only 2.8 per cent – less than a fifth of Caterpillar and the industry – and free cash flow was negative in three of four years. Our accounting tests put the margin even lower, between 1.6 and 2.8 per cent.

So our advice: do not invest now, and reassess on the 2026 statements. Let us show you why.

COMPANY AND ENVIRONMENT

Diversification kept Liebherr growing while construction weakened

€14,772 m	55,963	13	1949
Revenue 2025, +1.0 %	Employees	Product segments	Family-owned · Bulle (CH)

Revenue 2025 by sector

63.3 % · Construction & mining · −5.5 %	36.7 % · Other · +14.7 %

€9,345 m vs. €5,427 m – components alone grew 35.5 %



External pressures

- **Demand:** European construction output forecast to grow only about 2 % a year until 2028
- **Prices and costs:** US tariffs and low-cost competitors – reported by Liebherr for its cranes
- **Financing:** ECB main refinancing rate 2.65 % (Sept. 2026) – machines cost customers more to finance

LIEBHERR'S OWN OUTLOOK

Orders at the start of 2026 above the previous year – further growth expected, but **no significant upswing before 2027**. US tariff policy remains a risk.

Source: Report, chapters 2 and 3 (§3.1.1) and §5.7; Liebherr-International AG (2026); EUROCONSTRUCT (2026); European Central Bank (2026)

03

Liebherr is a family-owned, unlisted group, founded in 1949, with almost 56,000 employees in 13 product segments.

In 2025 construction and mining – almost two thirds of revenue – fell by 5.5 per cent, but the other segments grew by 14.7 per cent. That is why the group still grew a little.

The environment is difficult: slow European construction growth, US tariffs and low-cost competitors, and higher interest rates for customers. Liebherr itself expects no real upswing before 2027. So how does Liebherr compete in this market?

04 Competitive advantage

1:00

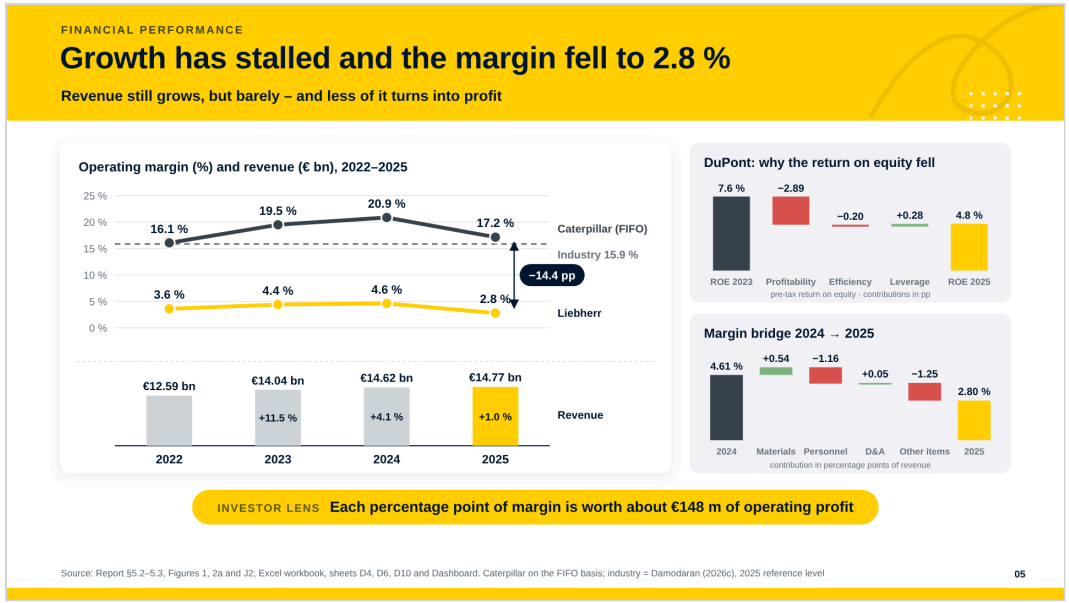


Liebherr competes with differentiation in custom businesses like cranes and mining, and with scale in volume businesses like earthmoving. Three advantages support this.

First, diversification: 13 segments spread the risk – in 2025 the other segments grew while construction fell. Second, vertical integration: Liebherr builds key technologies itself and sells through its own companies, not through dealers. Third, innovation: R&D rose every year to 708 million euro, 4.8 per cent of revenue – more than Caterpillar.

But each advantage has a price: diversification does not raise the margin, vertical integration ties up capital, and R&D kept rising while profit fell.

So the advantage is real, but it sits in the asset base, not yet in the margin. Does it turn into returns? The numbers show it.



The bars at the bottom show revenue: it grew 11.5 per cent in 2023, but only 1 per cent in 2025.

The lines above show the bigger problem. Liebherr's margin, in yellow, was around 4 per cent and fell to 2.8 per cent in 2025. Caterpillar earned 16 to 21 per cent, the industry 15.9. The gap existed in every single year – so this is not one bad year.

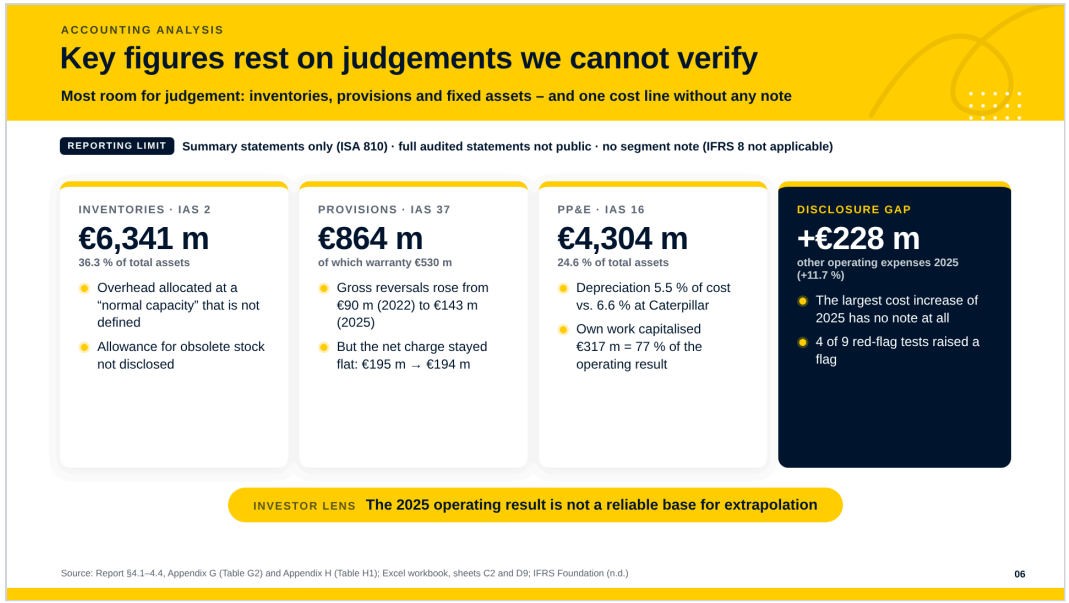
The bridge on the right explains 2025: personnel costs grew five times faster than revenue and cost 1.2 points, and other operating items cost another 1.25 points – without any explanation.

The DuPont analysis shows what this means for the owners: the pre-tax return on equity fell from 7.6 to 4.8 per cent, and almost all of it comes from lower profitability – not from efficiency or debt.

Each margin point is worth about 148 million euro. But can we trust the reported margin?

06 Accounting analysis – judgements

0:55



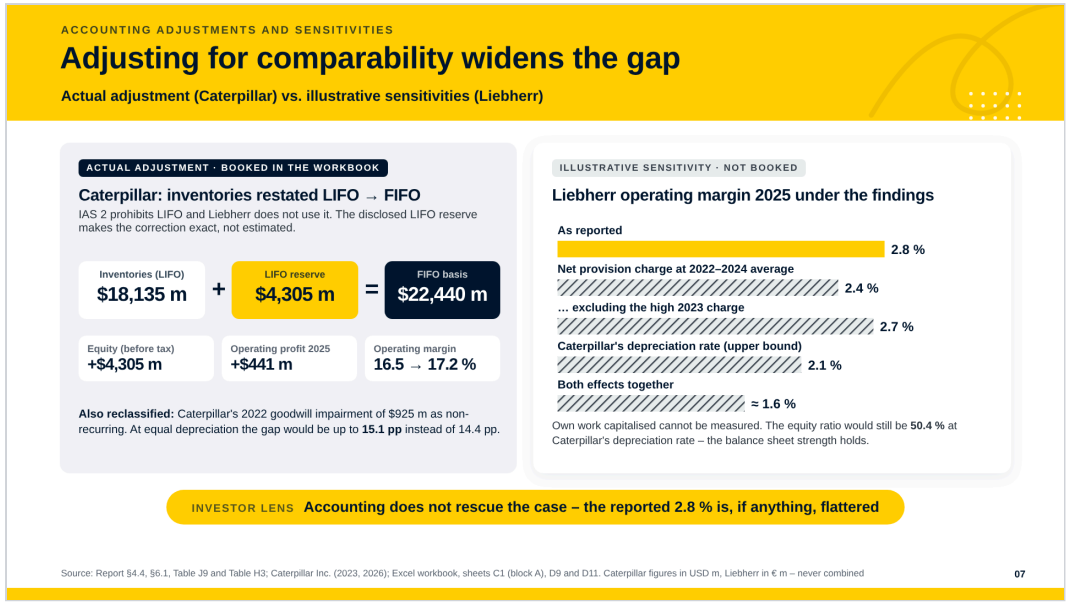
Liebherr publishes only summary statements – the full audited statements are not public, and there is no segment note.

Three accounts carry most of the judgement. Inventories: overhead is allocated at a normal capacity that is not defined, and the allowance for old stock is not disclosed. Provisions: reversals rose to 143 million euro, but the net charge stayed flat. Fixed assets: Liebherr depreciates more slowly than Caterpillar, and own work capitalised equals 77 per cent of the operating result.

The fourth box matters most: other operating expenses rose by 228 million euro, and there is no note at all. So the biggest cost increase of 2025 cannot be explained. Four of our nine red-flag tests raised a flag.

07 Accounting adjustments vs. sensitivities

1:00



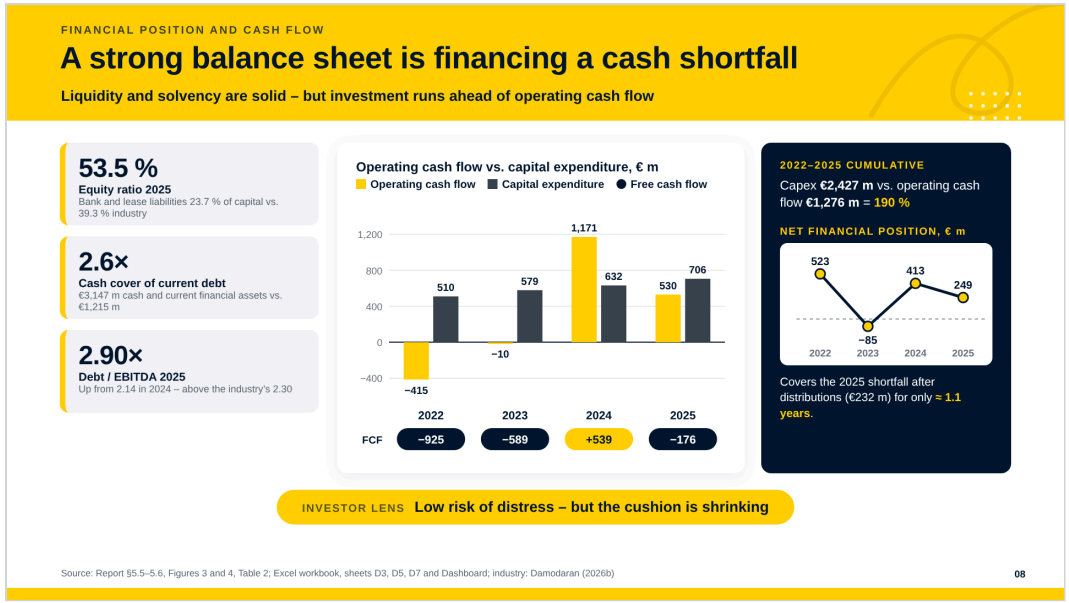
On the left is an actual adjustment we made; on the right are sensitivities we only measured.

Caterpillar values most inventory with LIFO, which IAS 2 does not allow and Liebherr does not use. Caterpillar discloses its LIFO reserve, so the correction is exact: plus 4.3 billion dollars on inventories and equity, plus 441 million on operating profit. Its margin rises from 16.5 to 17.2 per cent – so the gap gets bigger, not smaller.

For Liebherr we only illustrate: with a normal provision charge the margin would be 2.4 per cent, at Caterpillar's depreciation rate 2.1, and both together about 1.6 per cent.

So the accounting does not hide a better company. But the balance sheet strength survives: the equity ratio stays around 50 per cent.

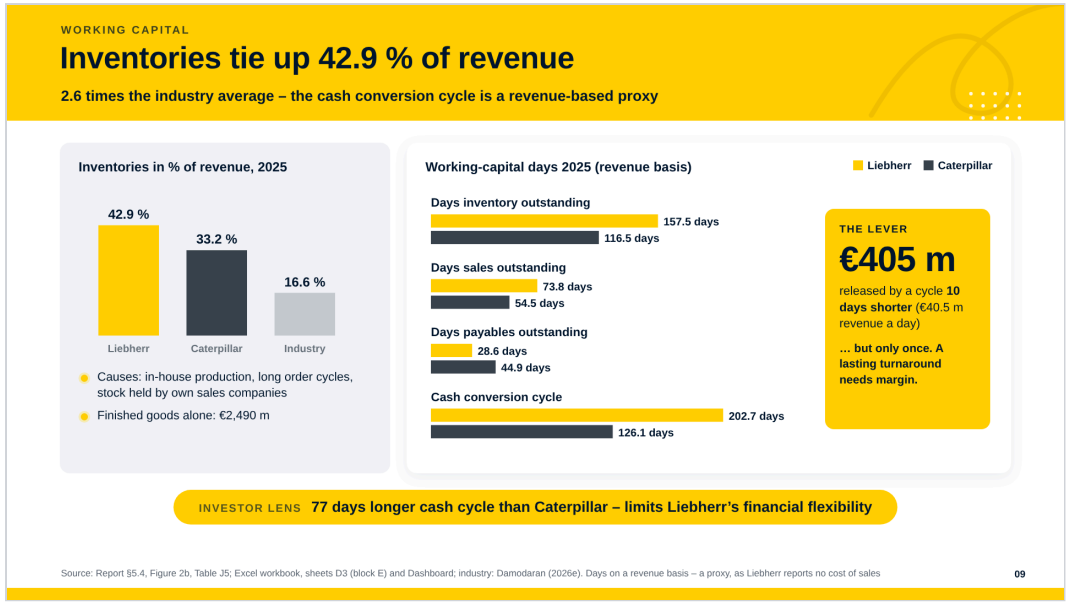
08 Financial position and cash flow



The balance sheet is Liebherr's clearest strength: 53.5 per cent equity, much less debt than the industry, and cash covering current debt 2.6 times.

But look at the cash. Yellow is operating cash flow, grey is capital expenditure. In 2022 and 2023 operating cash flow was even negative because Liebherr built up stock. Over four years it invested 2.4 billion euro but generated only 1.3 billion from operations – 190 per cent. Free cash flow was negative in three of four years.

So the cushion shrinks: the net financial position fell from 523 to 249 million euro, and debt to EBITDA rose to 2.9, above the industry. If the 2025 shortfall repeated, the cushion would last only about 1.1 years. The balance sheet buys time – it does not replace returns.



Where is the cash tied up? Mainly in inventories: 42.9 per cent of revenue, 2.6 times the industry.

Measured in days of revenue – a proxy, because Liebherr reports no cost of sales – the cash conversion cycle is about 203 days, 77 days longer than Caterpillar. Liebherr holds stock longer and pays suppliers faster.

This is the biggest lever management controls: ten days less would release about 405 million euro. But only once – so it cannot replace a better margin.

OVERALL ASSESSMENT AND INVESTMENT PERSPECTIVE

What would have to change: three conditions for a reassessment

Check them on the 2026 statements – a new assessment only if all three are met. In 2025 none of them is.



1 **Operating margin of at least 3.9 % (four-year average)**	2 **Free cash flow positive and, over time, covering distributions**	3 **Information rights: audited statements and other operating expenses**
Status 2025 × 2.8 %	Status 2025 × −€176 m	Status 2025 × Not secured

REWARD

Resilient, diversified group · back to the 3.9 % average margin = **+€157 m** operating profit at 2025 revenue

RISK AND LIMITS

ROCE 3.5 % (pre-tax) vs. 7.70 % benchmark – needs a margin of ~6–8 % · minority stake: little influence, low payout, hard to sell

Even then: meeting the conditions triggers a new analysis, not a purchase – a minority stake in an unlisted group would only make sense at a discounted price.

Source: Report §5.3, §5.7 and chapter 8, Table 3; Excel workbook, sheets D3, D10 and D11; benchmark: Damodaran (2026a)

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So what would have to change before an investment makes sense? We check three conditions on the 2026 statements.

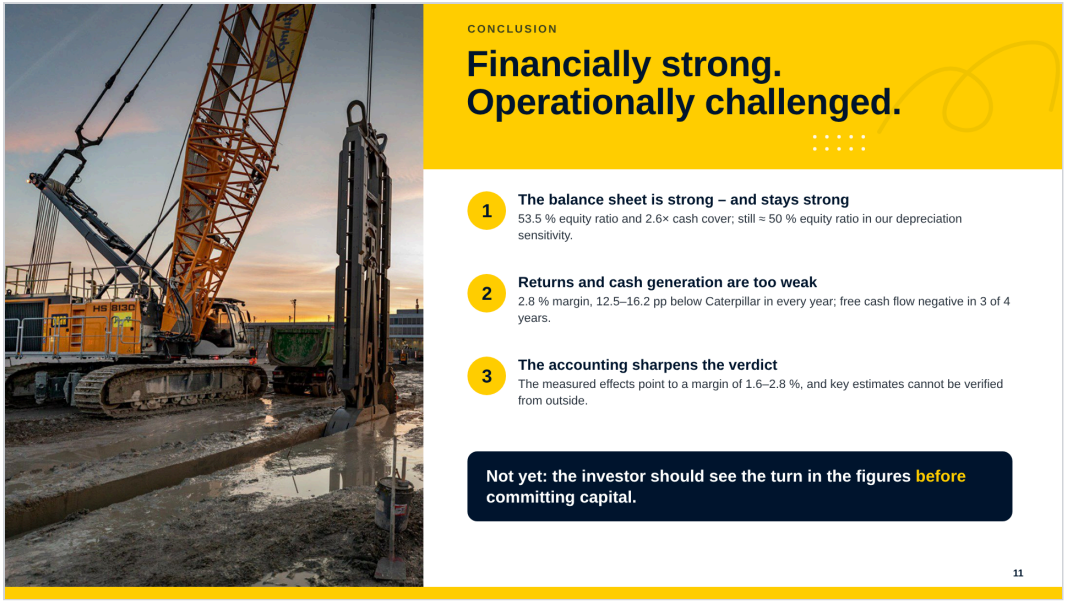
First, the margin should return to at least its four-year average of 3.9 per cent – in 2025 it was 2.8. Second, free cash flow should be positive and cover distributions – it was minus 176 million. Third, the investor needs information rights, for example on the 228 million euro of unexplained other costs.

And 3.9 per cent is only a trigger, not a good return: the return on capital employed was 3.5 per cent before tax, against a 7.7 per cent benchmark.

The reward is real – the average margin alone adds 157 million euro of profit. But a minority shareholder has little influence, a low payout and shares that are hard to sell. So meeting the conditions starts a new analysis, and the price would need a discount.

11 Conclusion

0:40



So, our answer to the central question: Liebherr is financially strong, but operationally challenged.

Three takeaways. The balance sheet is strong and survives our accounting tests. Returns and cash generation are too weak, and the gap to Caterpillar exists in every year. And the accounting analysis sharpens the verdict rather than softening it.

If the margin stays near 2.8 per cent in 2026 and the net financial position turns negative, the investor should drop the case. In one sentence: see the turn in the figures before committing capital.

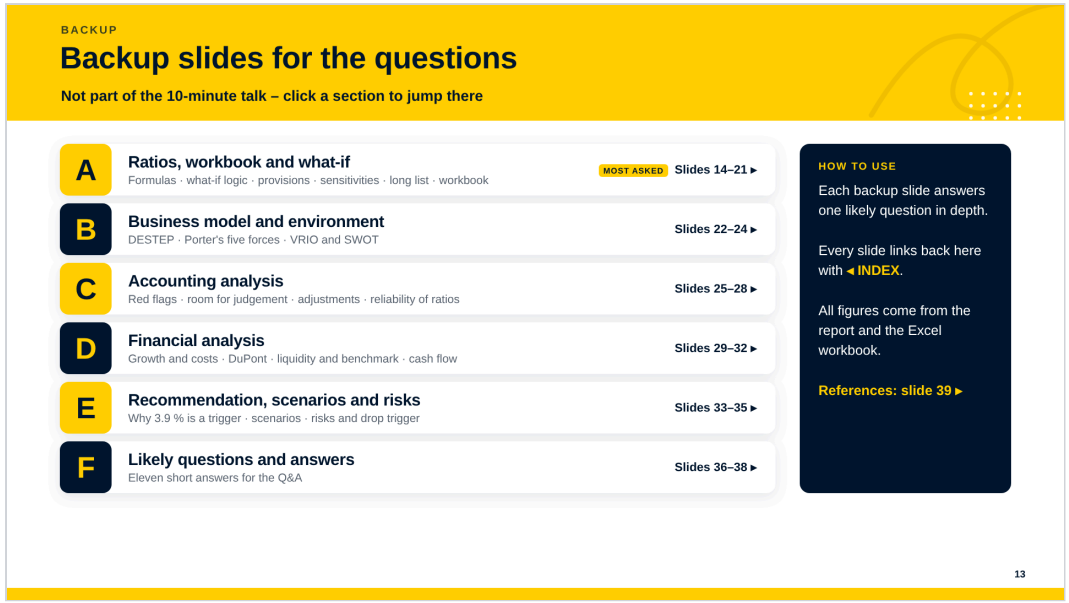
12 Thank you



Thank you for your attention. We are happy to answer your questions.

Back-up figures for Q&A: ROCE 2024 5.9 % before tax at a 4.6 % margin; sustainable growth rate 2.4 %; personnel expenses 28.6 % of revenue (26.4 % in 2023); only €56 m of a €272 m profit distributed in 2025; a deferred tax liability on Caterpillar's LIFO reserve at 21 % would lower its equity ratio only from 24.9 % to 24.2 %.

13 Backup index



Backup slides, not part of the 10-minute talk. Use them to answer questions: A covers ratios, formulas, what-if logic, provisions, sensitivities, the long and short list and the workbook – the most likely questions. B covers the business model, C the accounting analysis, D the financial analysis, E the recommendation and scenarios, F likely questions. Click a section to jump there; every backup slide links back to this index.

BACKUP · A | RATIOS AND WORKBOOK

Ratios I: returns are low because the margin is low

◂ INDEX

Profitability and returns – formula, Liebherr and Caterpillar, 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
PROFITABILITY	Operating margin	EBIT / revenue	2.8 %	17.2 %
	Pre-tax margin	Profit before tax / revenue	3.0 %	17.9 %
	Net profit margin	Profit after tax / revenue	1.8 %	13.8 %
	EBITDA margin	EBITDA / revenue	6.8 %	19.8 %
RETURNS	ROA	EBIT / average total assets	2.4 %	11.9 %
	ROCE	EBIT / (avg. equity + avg. non-current debt)	3.5 %	20.2 %
	ROE	Profit after tax / average equity	3.0 %	38.1 %*
	Pre-tax ROE	Profit before tax / average equity	4.8 %	49.4 %

* Not comparable – treasury shares reduce Caterpillar's equity. ROA is measured on EBIT, because Liebherr's interest expense is not disclosed.

DUPONT

ROE = net margin × asset turnover × leverage

leverage = average total assets / average equity

	MARGIN	TURNOVER	LEVERAGE	ROE
2025	1.84 %	× 0.85	× 1.89	3.0 %
pre-tax	2.96 %	× 0.85	× 1.89	4.8 %
2023 pre-tax	4.76 %	× 0.88	× 1.81	7.6 %

Leverage works on ROE, **not on the EBIT margin** – EBIT is measured before interest.

Source: Excel workbook, sheets D3 (blocks C and D) and D4; Report §5.3, Table J14. 2025; Caterpillar on the FIFO basis

Return on equity is profit after tax divided by average equity: 272 over 9,128 is 3.0 per cent. Before tax it is 4.8 per cent. DuPont splits it into net margin, asset turnover and leverage, where leverage is average total assets over average equity. Leverage works on return on equity, not on the EBIT margin, because EBIT is measured before interest. ROA is measured on EBIT because Liebherr does not disclose its interest expense. Caterpillar's return on equity is inflated by its treasury shares and is not comparable.

Backup A – Ratios II

BACKUP · A | RATIOS AND WORKBOOK

Ratios II: strong liquidity, slow capital

Asset utilisation, working capital, liquidity and solvency – 2025

◂ INDEX

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
ASSET UTILISATION	Asset turnover	Revenue / average total assets	0.85	0.69
	Fixed asset turnover	Revenue / average PP&E	3.49	4.74
WORKING CAPITAL	Days inventory outstanding	Avg. inventories / revenue × 365	157.5	116.5
	Days sales outstanding	Avg. receivables / revenue × 365	73.8	54.5
	Days payables outstanding	Avg. payables / revenue × 365	28.6	44.9
	Cash conversion cycle	DIO + DSO - DPO	202.7	126.1
LIQUIDITY	Current ratio	Current assets / current liabilities	2.31	1.55
	Quick ratio	(Current assets - inventories) / current liab.	1.16	0.94
	Cash cover	Cash + current fin. assets / current debt	2.6×	–
SOLVENCY	Equity ratio	Equity / total assets	53.5 %	24.9 %
	Debt / equity	Debt incl. pensions / equity	0.40	1.84
	Debt / capital	Bank + lease liab. / (those + equity)	23.7 %	64.8 %
	Debt / EBITDA	Bank + lease liabilities / EBITDA	2.90	3.52

READ WITH CARE

Days on a revenue basis: Liebherr reports no cost of sales. On its own cost basis DIO would be 299 days – not comparable.

Two debt measures: "debt" includes pensions (€3,714 m); bank and lease liabilities (€2,898 m) match the Damodaran benchmark.

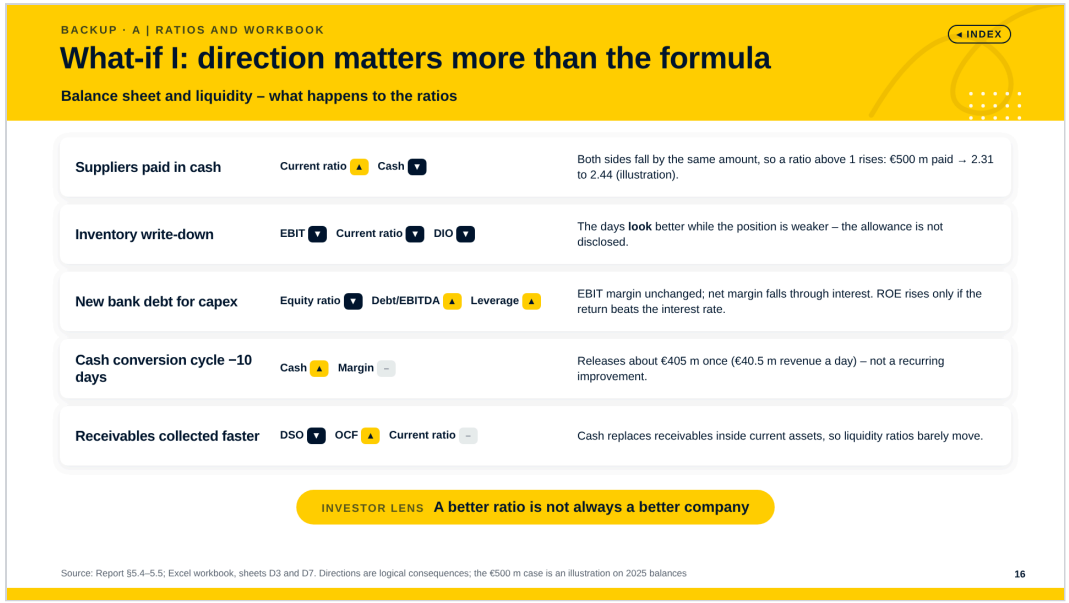
Interest cover: not computable – no interest expense disclosed. Indicator: operating profit / interest paid 3.90.

Source: Excel workbook, sheets D3 (blocks D, E, G, H) and D7; Report Tables J5 and J14. 2025; days on average balances; Caterpillar on the FIFO basis

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The current ratio and the quick ratio are liquidity ratios; equity ratio, debt to equity and debt to EBITDA are solvency ratios. The working-capital days are calculated on revenue, because Liebherr reports no cost of sales; on its own cost basis inventory days would be 299 and not comparable. There are two debt measures: debt including pensions of 3,714 million euro, which gives debt to equity of 0.40, and bank and lease liabilities of 2,898 million euro, which match the Damodaran benchmark and give 23.7 per cent of capital and 2.90 times EBITDA. Interest cover cannot be computed for Liebherr.

Backup A – What-if I



BACKUP · A | RATIOS AND WORKBOOK

What-if I: direction matters more than the formula

◂ INDEX

Balance sheet and liquidity – what happens to the ratios

Suppliers paid in cash	Current ratio ▲ Cash ▼	Both sides fall by the same amount, so a ratio above 1 rises: €500 m paid → 2.31 to 2.44 (illustration).
Inventory write-down	EBIT ▼ Current ratio ▼ DIO ▼	The days **look** better while the position is weaker – the allowance is not disclosed.
New bank debt for capex	Equity ratio ▼ Debt/EBITDA ▲ Leverage ▲	EBIT margin unchanged; net margin falls through interest. ROE rises only if the return beats the interest rate.
Cash conversion cycle −10 days	Cash ▲ Margin –	Releases about €405 m once (€40.5 m revenue a day) – not a recurring improvement.
Receivables collected faster	DSO ▼ OCF ▲ Current ratio –	Cash replaces receivables inside current assets, so liquidity ratios barely move.

INVESTOR LENS **A better ratio is not always a better company**

Source: Report §5.4–5.5; Excel workbook, sheets D3 and D7. Directions are logical consequences; the €500 m case is an illustration on 2025 balances

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If suppliers are paid in cash, current assets and current liabilities fall by the same amount, so a current ratio above one rises – with 500 million euro it would go from 2.31 to 2.44. An inventory write-down lowers profit and the current ratio, but also the inventory days, so the ratio looks better while the position is weaker. New debt lowers the equity ratio and raises leverage, but leaves the EBIT margin unchanged. A shorter cash conversion cycle releases cash once, it does not improve the margin.

Backup A – What-if II

BACKUP · A | RATIOS AND WORKBOOK

What-if II: accounting choices move profit, not cash

◂ INDEX

Effect of an accounting choice on EBIT, equity and cash

CHANGE	EBIT	EQUITY	CASH / OCF	LIEBHERR EXAMPLE
Provision addition	▼	▼	–	Expense now – the obligation is estimated
Provision usage (utilisation)	–	–	▼	Obligation settled – no profit effect
Provision reversal (release)	▲	▲	–	Estimate no longer needed – profit without cash
Slower depreciation (longer life)	▲	▲	–	Liebherr 5.5 % vs. Caterpillar 6.6 % of cost
More own work capitalised	▲ now	▲	–	€317 m = 77 % of 2025 EBIT; returns as depreciation
LIFO → FIFO, rising prices (CAT)	▲	▲	–	+$441 m EBIT, +$4,305 m equity before tax

INVESTOR LENS **Operating cash flow is the check on the accruals – only 28.4 % of EBITDA in 2022–2025**

Source: Report §4.2–4.4 and §6.2; Excel workbook, sheets A2 (note 10), C1, C2 and D9

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Accounting choices move profit, not cash. A provision addition is an expense, usage settles the obligation in cash without a profit effect, and a reversal increases profit without cash. Slower depreciation and more own work capitalised raise EBIT today. Caterpillar's switch from LIFO to FIFO raises profit and equity. That is why operating cash flow, only 28.4 per cent of EBITDA over four years, is our check on the accruals.

Backup A – Provisions

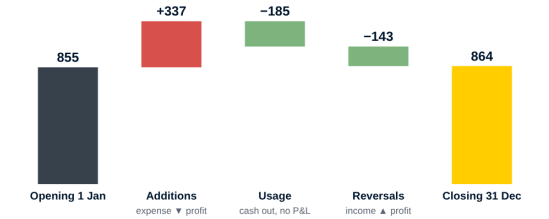
BACKUP · A | RATIOS AND WORKBOOK

Provisions: reversals rose – but so did additions

◂ INDEX

Movement of provisions 2025 (IAS 37), € m – of which warranty 530

Roll-forward 2025, all provisions



Net charge = additions − reversals	2022	2023	2024	2025
	193	338	195	194

How to read it

- **IAS 37:** present obligation from a past event, probable outflow, reliable estimate
- **Usage ≠ reversal:** only additions and reversals touch profit
- **Hidden reserve test:** a release to support profit would show as a *falling* net charge – 195 → 194, so not visible
- **Flag kept:** other provisions built up by €146 m in 2023 are being released (€63 m in 2025); the notes cannot show why

Source: Excel workbook, sheets A2 (note 10, rows 68–80), C2 and D11 (block B); Report §4.2 and §4.4, Table H1 test 1. € m

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Provisions under IAS 37 need a present obligation from a past event, a probable outflow and a reliable estimate. In 2025 the opening balance of 855 million euro rose by 337 of additions, fell by 185 of usage and by 143 of reversals, to 864. Usage is the real outflow and has no profit effect; only additions and reversals touch profit. The net charge – additions less reversals – stayed at 194 against 195, so a hidden reserve being released to support profit is not visible. We keep the flag because other provisions built up in 2023 are now being released and the notes cannot explain why.

BACKUP · A | RATIOS AND WORKBOOK

◂ INDEX

Sensitivity: every measured effect lowers the margin

Liebherr operating margin 2025 – illustrations on published figures, not booked adjustments

STEP	CALCULATION, € M	EBIT	MARGIN
As reported	–	413	2.80 %
Net provision charge at the 2022–2024 average	413 - (1.76 % - 1.31 %) × 14,772	347	2.35 %
… without the high charge of 2023	413 - (1.43 % - 1.31 %) × 14,772	395	2.68 %
Caterpillar's depreciation rate (6.6 % instead of 5.5 % of cost)	413 - 105	308	2.08 %
Both effects together	413 - 66 - 105	242	1.64 %

2.80 − 0.44 − 0.71 = 1.64 %

Sheet D9 also shows a **gross-reversal** view (2.5 % at average reversals, 1.8 % without any reversal). The report uses the **net charge** (D11), because additions rose together with the reversals. Own work capitalised (€317 m) cannot be measured.

Source: Excel workbook, sheets D9 (blocks A and B) and D11 (block B); Report §6.1, Tables H3 and J13

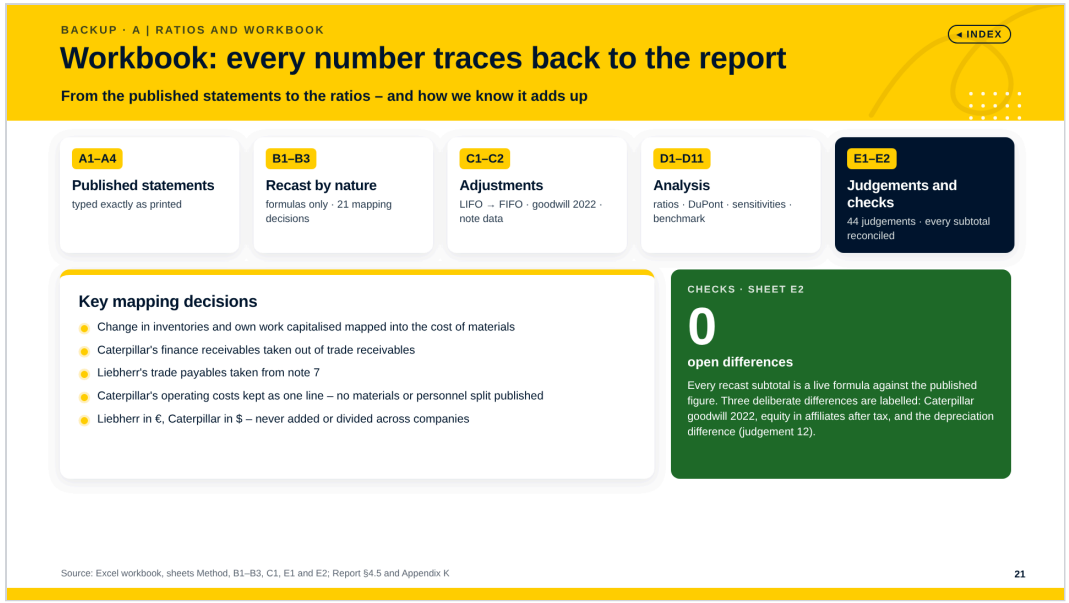
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We measure two effects without booking them. If the net provision charge had been at its 2022 to 2024 average of 1.76 per cent of revenue instead of 1.31, EBIT would be 347 million euro and the margin 2.35 per cent; without the high charge of 2023 it would be 2.68. At Caterpillar's depreciation rate depreciation would be 105 million euro higher and the margin 2.08 per cent. Both together give 242 million euro, or 1.64 per cent. Sheet D9 also shows a gross-reversal view, but the report uses the net charge because additions rose with the reversals.

Backup A – Long list to short list



Step one is the long list of 30 accounting treatments with room for choice. Step two keeps the nine that combine a key account with high or medium-to-high flexibility. Step three is the red-flag screen: eight tests plus one consistency check, with four flags, three clean and two not testable. Step four is the short list of 22 findings: seven that no adjustment can remove, five reversed in the workbook and ten the disclosure does not allow us to reverse. The two counts of nine are a coincidence.

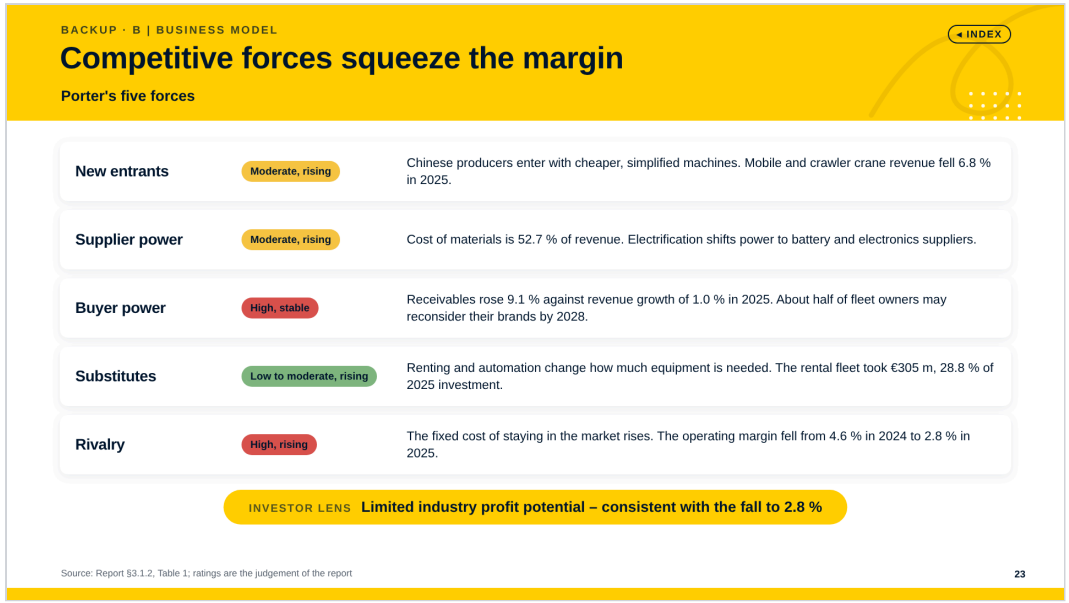


The workbook starts with the published statements typed exactly as printed. The B sheets recast both companies into one format by nature using formulas only, with 21 documented mapping decisions. The C sheets hold the LIFO to FIFO restatement and the note data, the D sheets the analysis. Sheet E1 documents 44 judgements and sheet E2 reconciles every recast subtotal with the published figure: zero open differences, apart from three labelled ones.

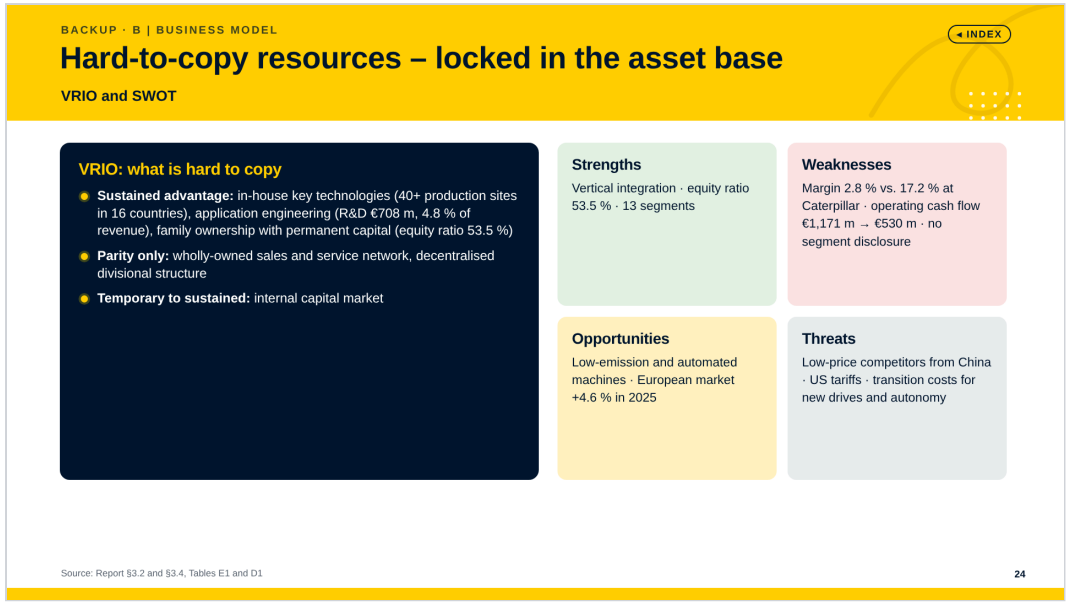


The DESTEP scan shows a modest European recovery from 2026, set against higher financing costs, tariff uncertainty and a technology and labour transition that raises costs before it raises prices. Economic and political factors carry the largest near-term risk, technological and ecological factors the largest long-term opportunity. A market recovery alone does not show that the margin recovers.

23 Backup B – Five forces



Buyer power and rivalry are high, the threat of new entrants and supplier power are moderate and rising, and substitutes are a low to moderate threat. The industry profit potential is therefore limited and squeezed from the customer side and by low-cost entrants. This is consistent with the fall of the operating margin to 2.8 per cent in 2025 against 17.2 per cent at Caterpillar.



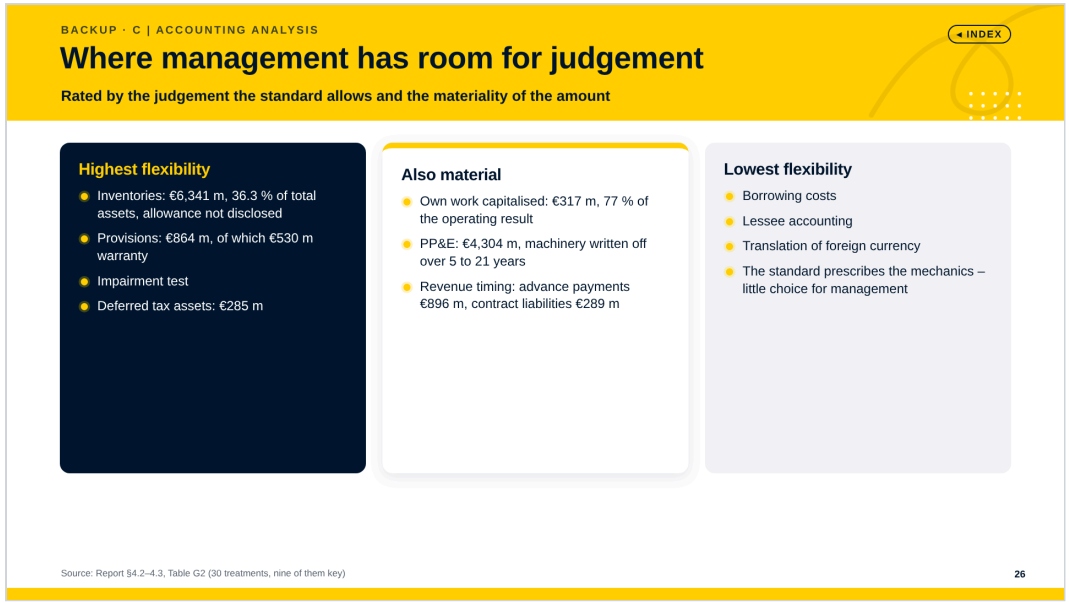
The VRIO test shows that in-house technology, application engineering and family capital are sustained advantages, while the sales and service network is only at parity, because Caterpillar's independent dealers provide comparable access. The SWOT sums it up: vertical integration and the equity ratio are the strengths that matter, the weakness is the margin and the weakening internal funding, the opportunity is demand for low-emission and automated machines, and the threats are low-price competitors and tariffs.

Backup C – Red flags



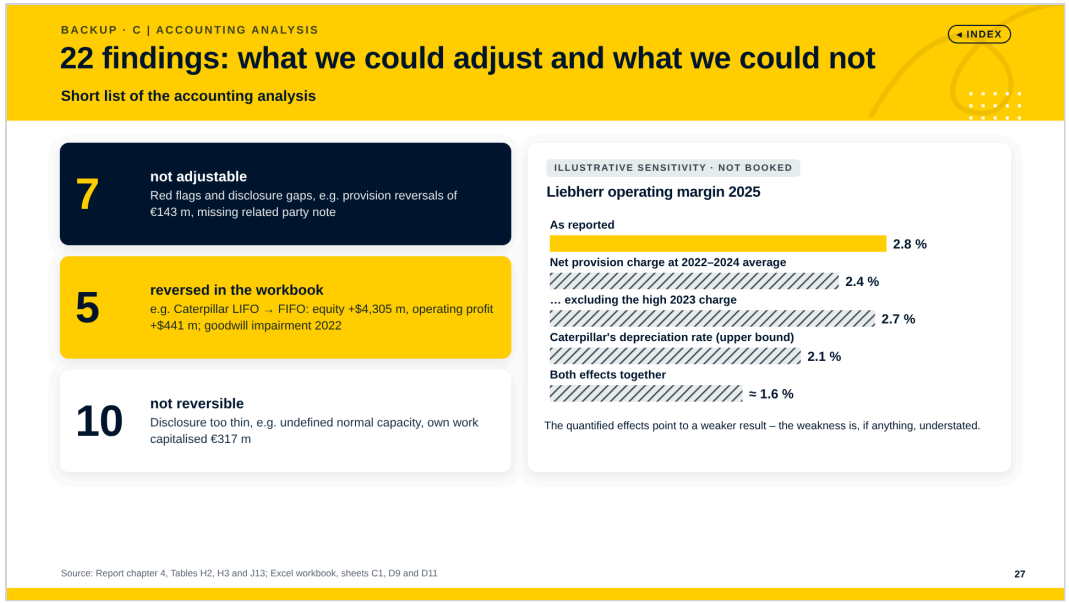
We ran nine red-flag tests over four years. Four raised a flag, three came back clean and two cannot be run. The flags are the provision reversals of 143 million euro, receivables growing faster than revenue, an effective tax rate between 37.8 and 99.2 per cent without a reconciliation, and the impairment of 55 million euro in 2023. The tests that cannot be run are related party transactions and year-end adjustments. Red flags are reasons to ask, not proof of manipulation.

Backup C – Room for judgement



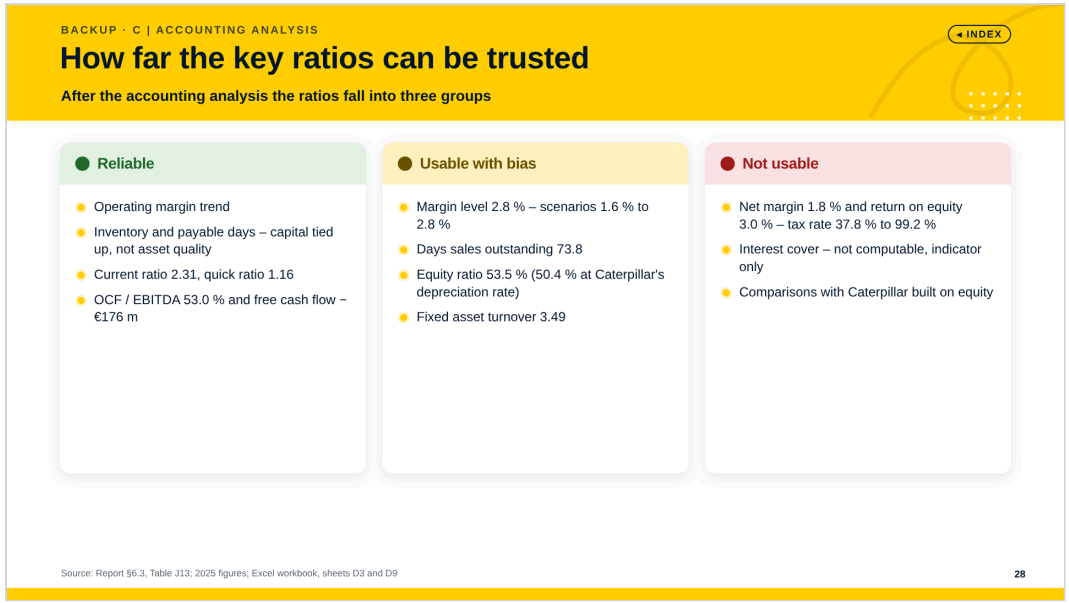
Flexibility is rated on two criteria: the degree of judgement the standard permits and the materiality of the amount. It is highest for inventories and their allowance, the impairment test, the provisions and deferred tax assets. Own work capitalised, property, plant and equipment and revenue timing follow. It is lowest where the standard prescribes the mechanics, as with borrowing costs, lessee accounting and currency translation.

27 Backup C – Adjustments and limits



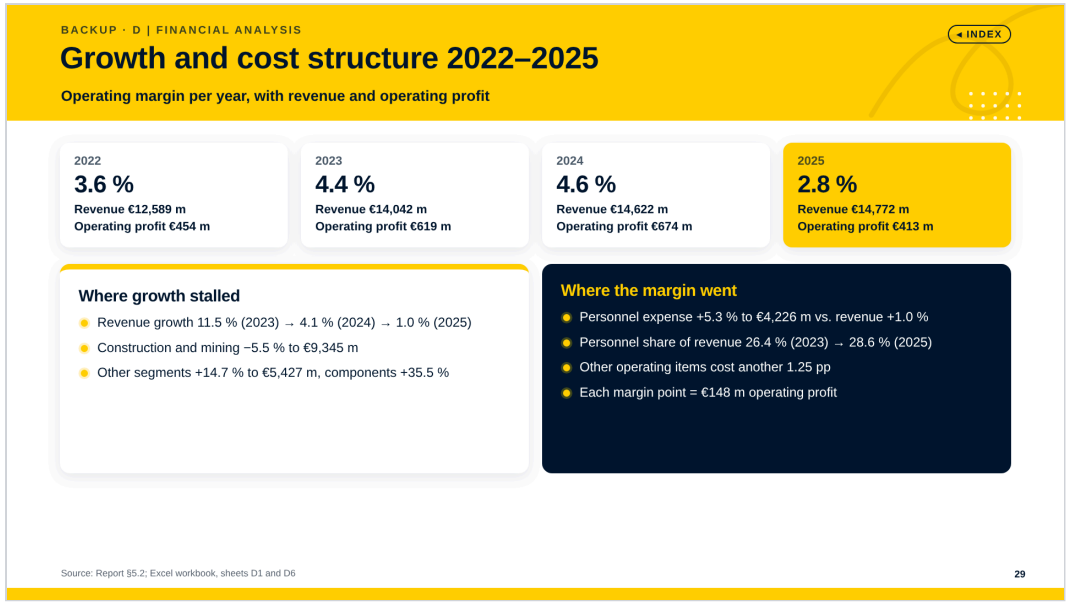
The short list holds 22 accounting findings. Seven are red flags or disclosure gaps that no adjustment can remove, five were reversed in the workbook, and ten cannot be reversed because the disclosure is too thin. The quantified effects make the 2025 result look better, not worse: the operating margin of 2.8 per cent falls to between about 1.6 and 2.7 per cent in the measured scenarios, so the weakness is, if anything, understated.

28 Backup C – Reliability of ratios



After the accounting analysis the ratios fall into three groups. Reliable are the trend of the operating margin, inventory and payable days, liquidity and cash conversion. Usable with a stated bias are the level of the margin, days sales outstanding, fixed asset turnover and the equity ratio. Not usable for a conclusion are the net margin and the return on equity as a trend, interest cover and every comparison with Caterpillar built on equity.

29 Backup D – Growth and costs



Revenue growth slowed from 11.5 per cent in 2023 to 1.0 per cent in 2025. The slowdown came from construction and mining, which fell 5.5 per cent, while the other segments grew 14.7 per cent. The margin fell from 4.6 to 2.8 per cent because personnel expense grew 5.3 per cent against revenue growth of 1.0 per cent, and other operating items added to the burden. Each margin point is worth 148 million euro.

Backup D – DuPont

BACKUP · D | FINANCIAL ANALYSIS

DuPont: the margin, not leverage, drives returns down

◂ INDEX

Pre-tax return on equity 2023 → 2025

Contributions to the change in pre-tax ROE, percentage points



What it means

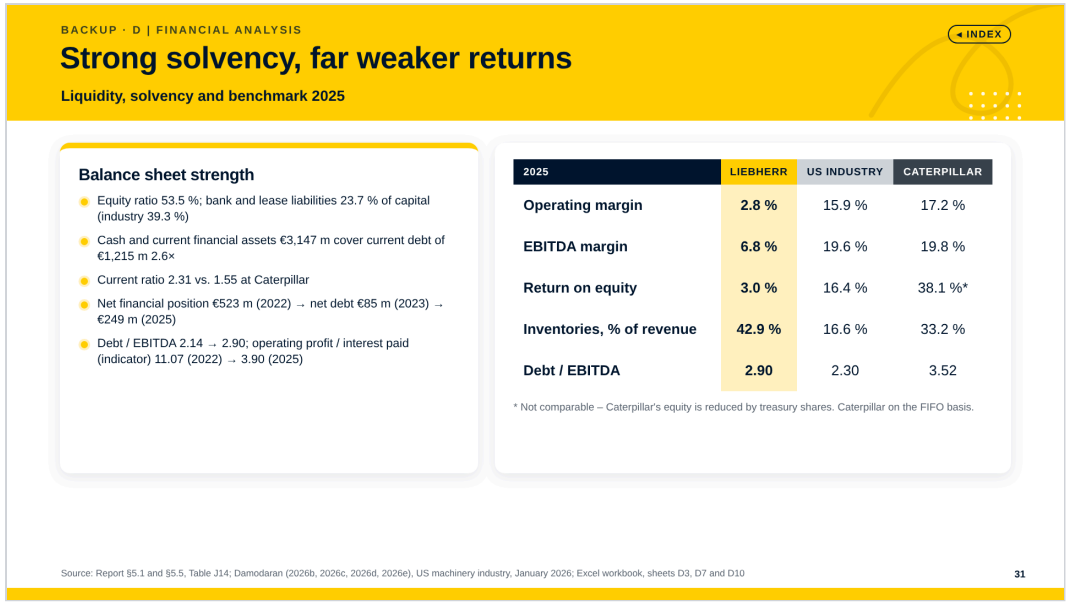
- **Profitability −2.89 pp:** lower pre-tax margin
- **Efficiency −0.20 pp:** slightly lower asset turnover
- **Leverage +0.28 pp:** more debt even helped a little
- After tax, ROE was 3.0 % vs. 16.4 % in the US industry – but the tax charge is volatile, so the trend is read before tax

Source: Report §5.3, Figure 2a, Table J4; Excel workbook, sheets D4 and D6 (logarithmic split)

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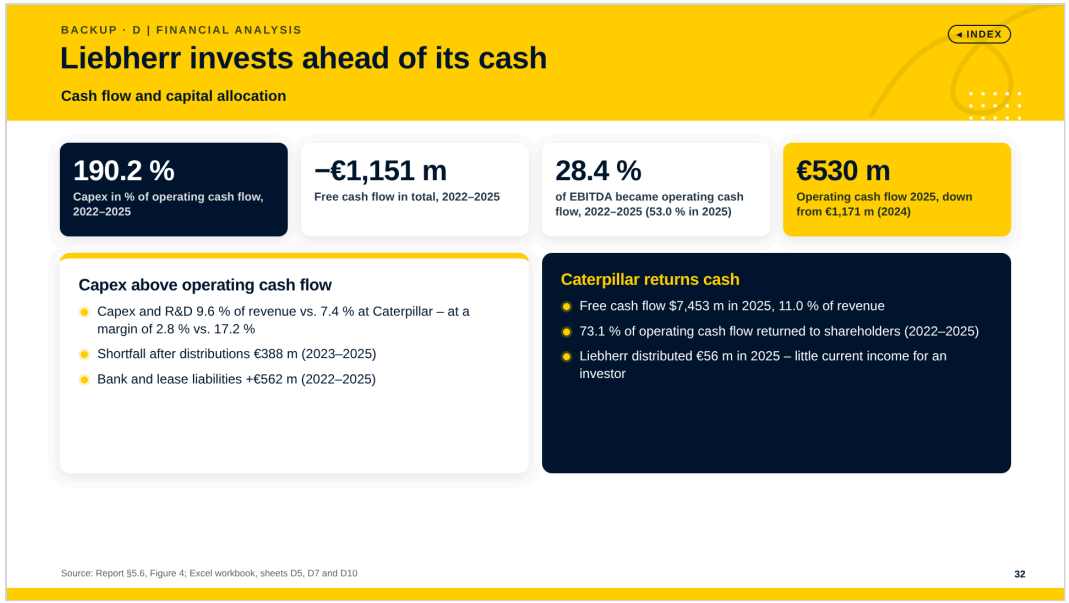
We read the return on equity before tax because the tax charge is volatile. The pre-tax return fell from 7.6 per cent in 2023 to 4.8 per cent in 2025. The DuPont decomposition assigns 2.89 percentage points of the fall to the lower pre-tax margin and 0.20 points to lower asset turnover, while higher leverage added 0.28 points. Leverage is therefore not the cause – the margin is.

Backup D – Liquidity and benchmark



Solvency is the clearest strength, with an equity ratio of 53.5 per cent and a current ratio of 2.31. The cushion is shrinking, though: the net financial position fell from 523 million euro in 2022 to a net debt of 85 million in 2023 and stood at 249 million in 2025, and debt to EBITDA rose to 2.90, above the industry's 2.30. Against the industry and Caterpillar, margins and returns are far lower and inventories tie up much more capital. Caterpillar's return on equity is not comparable because of its treasury shares.

Backup D – Cash flow



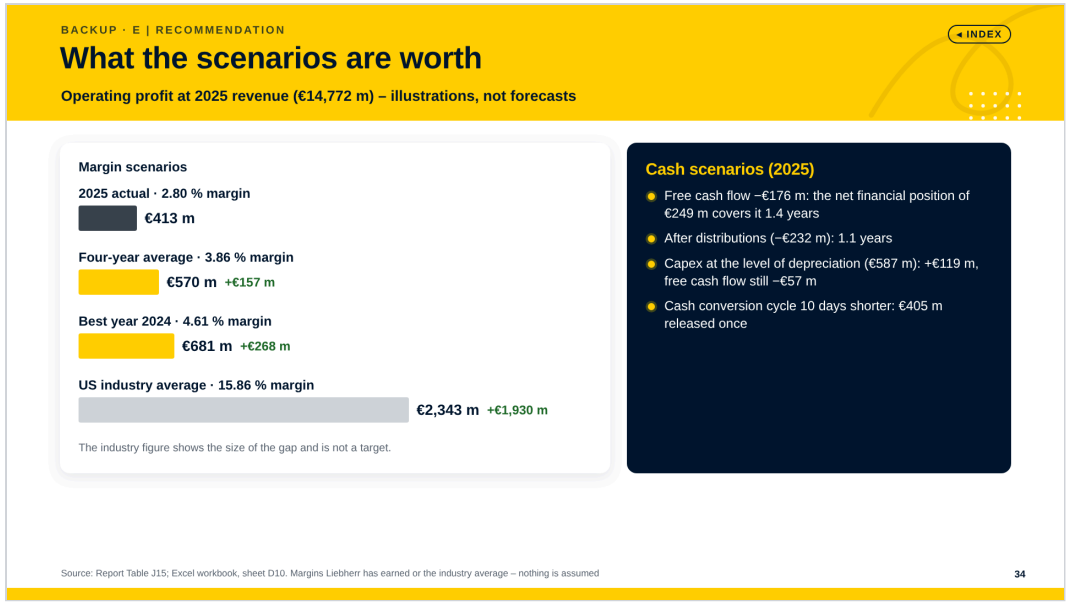
Cash flow is the weakest part of the figures. From 2022 to 2025 capital expenditure equalled 190.2 per cent of operating cash flow and free cash flow totalled minus 1,151 million euro. Operating cash flow fell from 1,171 million euro in 2024 to 530 million in 2025. Liebherr spends more than Caterpillar relative to revenue but earns a far lower margin, and borrowing helped fund investment and distributions. Caterpillar generated free cash flow of 7,453 million dollars and returned 73.1 per cent of its operating cash flow to shareholders.

Backup E – Why 3.9 %



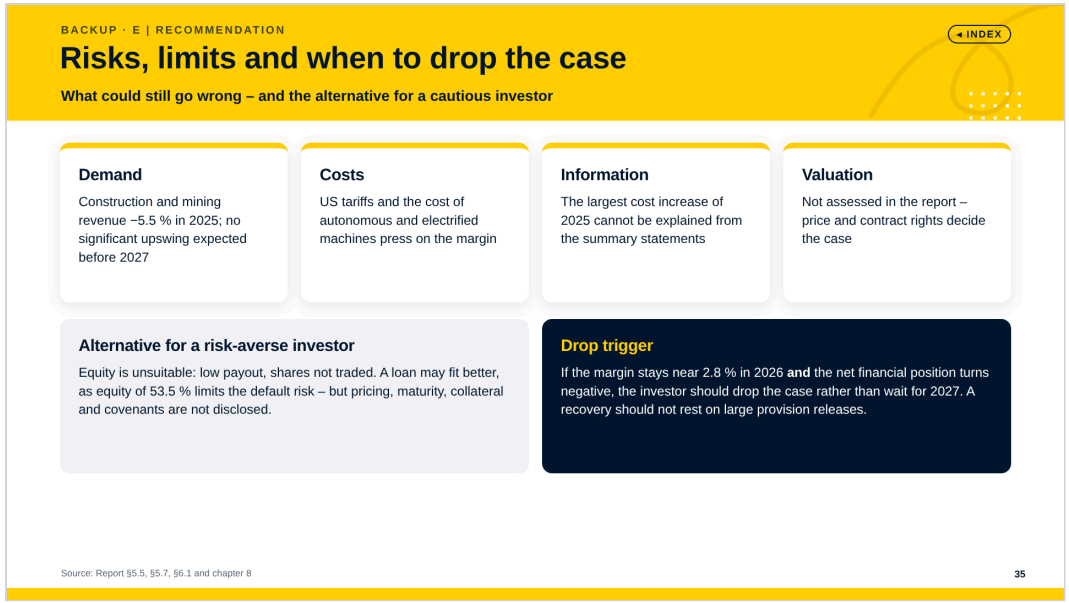
The margin of 3.9 per cent is the four-year average and therefore a realistic first test of recovery. At 2025 revenue it would add 157 million euro of operating profit. It is a trigger, not a target return: even in 2024, at a margin of 4.6 per cent, the return on capital employed was 5.9 per cent before tax, below the 7.70 per cent after-tax benchmark. Earning the benchmark would need a margin of about 6 to 8 per cent. The benchmark is that of listed US machinery companies, not Liebherr's own cost of capital.

Backup E – Scenarios



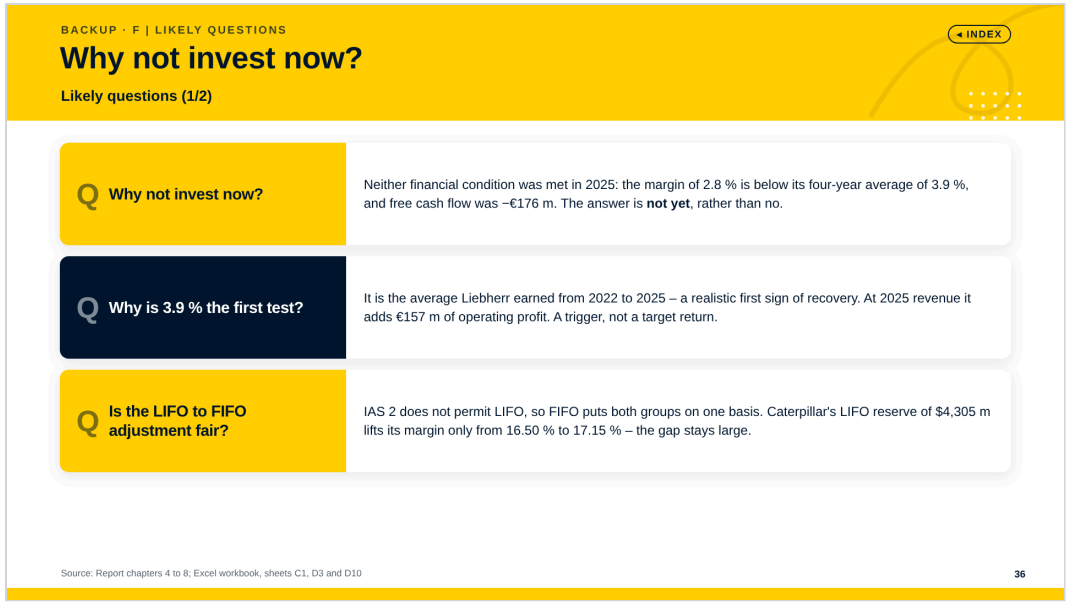
The margin scenarios apply margins Liebherr has already earned to the revenue of 2025 – illustrations on published figures, not forecasts. The industry margin shows the size of the gap and is not a target. On the cash side the net financial position of 249 million euro covers the free cash flow shortfall of 176 million euro for 1.4 years and the shortfall after distributions for 1.1 years. Capital expenditure at the level of depreciation would improve free cash flow by 119 million euro, but it would still be negative at minus 57 million. Ten fewer days in the cash conversion cycle would release about 405 million euro, once.

35 Backup E – Risks and drop trigger



Demand, tariffs and the cost of new technology are the main external risks. Financing is not the immediate constraint, but prolonged underperformance would make it one. The investor needs information rights because the largest cost increase of 2025 cannot be explained. The report does not assess a valuation. For a risk-averse investor an equity stake is unsuitable and a loan might fit better in principle, but its terms are not disclosed. The drop trigger is a margin near 2.8 per cent in 2026 together with a negative net financial position.

Backup F – Questions 1/2



Answer 1 is the recommendation in short. Answer 2 explains why the four-year average is only a first test. Answer 3 shows that the adjustment of Caterpillar to FIFO is small compared with the gap in margin.

Backup F – Questions 2/2

BACKUP · F | LIKELY QUESTIONS

What the balance sheet cannot fix

Likely questions (2/2)

◂ INDEX

Q What do the red flags mean?	They are reasons to ask, not proof of manipulation. Four flags are raised, three tests are clean and two cannot be run on the published statements – that is why information rights come first.
Q Why is ROCE so low?	ROCE is 3.5 % before tax against a 7.70 % after-tax benchmark: the margin is only 2.8 %, and inventories of 42.9 % of revenue and a cash cycle 77 days longer than Caterpillar's tie up capital.
Q Why is a strong balance sheet not a return?	An equity ratio of 53.5 % and a net financial position of €249 m limit the risk, but do not earn the cost of capital. They would cover a shortfall like 2025 (−€232 m) for about 1.1 years.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

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The red flags are reasons to ask. ROCE is low because of the low margin and the capital tied up in inventories. The balance sheet protects the investor, but it cannot replace better returns and cash conversion.

Backup F – Questions 3/3

BACKUP · F | LIKELY QUESTIONS

Questions on method

◂ INDEX

Likely questions (3/3)

Q Why pre-tax ROE?	The effective tax rate was 99.2 %, 45.1 %, 58.2 % and 37.8 % (2022–2025), with no tax reconciliation published – the after-tax trend says little.
Q Why Caterpillar as the peer?	The world's leading maker of construction and mining equipment – the core market of Liebherr's largest sector (63.3 % of revenue). We compare ratios only, never € and $ figures.
Q Why a US industry benchmark?	Damodaran publishes consistent data for 105 listed machinery companies. It is a reference level, not Liebherr's own cost of capital.
Q R&D expensed, yet €317 m capitalised?	Own work capitalised is self-built assets and rental machines (IAS 16), not development costs (IAS 38), which are never capitalised.
Q Why days on a revenue basis?	Liebherr reports no cost of sales, so DIO and DPO use revenue for both groups – comparable, but not the textbook formula.

Source: Report §4.2, §5.3, §5.7 and §6.3; Excel workbook, sheets C2 and D3; Caterpillar Inc. (2026)

38

We read returns before tax because the effective tax rate swung between 37.8 and 99.2 per cent without a reconciliation. Caterpillar is the world's leading maker of construction and mining equipment, the market of Liebherr's largest sector, and we only compare ratios. The Damodaran industry is a reference level, not Liebherr's cost of capital. Own work capitalised is self-built equipment, not development costs. Working-capital days use revenue because Liebherr reports no cost of sales.

BACKUP · REFERENCES

References

Sources cited in the presentation, including the backup slides (APA 7th edition)

◂ INDEX

Caterpillar Inc. (2023). *Annual report on Form 10-K for the fiscal year ended December 31, 2022*. U.S. Securities and Exchange Commission.

Caterpillar Inc. (2026). *Annual report 2025* [Including the annual report on Form 10-K for the fiscal year ended December 31, 2025]. U.S. Securities and Exchange Commission.

Committee for European Construction Equipment. (2026, February 26). *European construction equipment sector sees moderate recovery in 2025 with 4.6% growth in sales*.

Damodaran, A. (2026a). *Cost of equity and capital (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026b). *Debt fundamentals by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026c). *Margins by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026d). *Return on equity by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026e). *Working capital ratios by sector (US)* [Data set]. NYU Stern School of Business.

EUROCONSTRUCT. (2026, June 5). *Press release – 101st EUROCONSTRUCT conference* [Press release].

European Central Bank. (2026, September 10). *Monetary policy decisions* [Press release].

European Commission. (n.d.-a). *Emissions from non-road mobile machinery*.

European Commission. (n.d.-b). *European climate law*.

European Union. (2023). Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC of the European Parliament and of the Council and Council Directive 73/361/EEC. *Official Journal of the European Union*, L 165, 1–102.

Eurostat. (2025). *Accidents at work – statistics by economic activity*.

IFRS Foundation. (n.d.). *IFRS Accounting Standards Navigator*.

Liebherr-International AG. (2023). *Annual report 2022*.

Liebherr-International AG. (2024). *Annual report 2023*.

Liebherr-International AG. (2025). *Annual report 2024*.

Liebherr-International AG. (2026). *Annual report 2025*.

Own work: Beermann, S., Fliegel, A., Narkiewicz, Z., Hoffmann, P., & Gilmore Islam, Z. (2026). *Liebherr Group financial report: Investment analysis 2022–2025* and the Excel workbook *Liebherr–Caterpillar restructured financial statements*. Fontys University of Applied Sciences, Venlo. · Images: Liebherr-International AG. · Full entries with URLs: report, References.

Reference list for every source cited on the slides, in APA 7th edition. The full entries with URLs are in the report. Use this slide if someone asks where an external figure comes from.